
Classification Model Analysis with a Leukemia Dataset

Pedro. Silva¹

¹ *Tecnologico de Monterrey, Ingenieria en ciencia de datos y matematicas, Jalisco, Mexico*

Abstract— This study compares the performance and computational efficiency of two preprocessing pipelines for leukemia classification from blood smear images. A standard greyscale Convolutional Neural Network (CNN) pipeline is evaluated against a pipeline employing Principal Component Analysis (PCA) for dimensionality reduction to 50 features before classification with an identical CNN. The objective is to analyze the trade-off between model effectiveness and resource usage. Results indicate that while the greyscale CNN achieves superior accuracy, the PCA-based model trains significantly faster with a lower computational footprint. The findings demonstrate a direct trade-off, providing guidance for model selection based on available resources versus required diagnostic performance.

Keywords— Leukemia Classification, Convolutional Neural Networks, Principal Component Analysis, Computational Efficiency, Medical Image Analysis, Model Comparison

I. INTRODUCTION

The accurate and timely diagnosis of leukemia, a malignancy of the blood-forming tissues, is a critical component of effective treatment protocols. Traditional diagnosis relies on the microscopic examination of blood smears, a process that is inherently time-consuming and subject to inter-observer variability. Consequently, the automation of this analysis through deep learning has emerged as a significant research focus, with Convolutional Neural Networks (CNNs) demonstrating notable success in classifying medical images.

A primary challenge in deploying such models, particularly in resource-constrained clinical environments, is their substantial computational requirement. CNNs, while powerful, demand significant processing power and memory when trained on high-resolution image data. This limitation motivates the investigation of preprocessing methodologies that can streamline the classification pipeline. Dimensionality reduction techniques, such as Principal Component Analysis (PCA), present a promising approach by projecting image data into a lower-dimensional feature space, thereby reducing the input complexity for subsequent models.

This study conducts a comparative analysis of two distinct preprocessing pipelines for CNN-based leukemia classification. The first pipeline utilizes a standard greyscale conversion of the input images, preserving spatial structure for the CNN. The second pipeline employs PCA to reduce each image to 50 principal components before classification with an identical CNN architecture. The central hypothesis is that the greyscale CNN will achieve

higher classification performance by leveraging richer image data, whereas the PCA-CNN pipeline will demonstrate superior computational efficiency at the potential cost of diagnostic accuracy. The investigation quantitatively evaluates this trade-off, providing insights into the balance between model efficacy and practical deployability in medical imaging applications.

II. METHODOLOGY

This study employs a comparative experimental design to evaluate the trade-off between computational efficiency and classification performance in two distinct preprocessing pipelines for leukemia classification. The methodology encompasses data preparation, preprocessing techniques, model architecture, training procedures, and evaluation metrics, ensuring a controlled and reproducible comparison between conventional image processing and dimensionality reduction approaches.

a. Dataset Description and Preparation

The experimental framework utilized the publicly available Leukemia Classification dataset from Kaggle, comprising 15,135 high-resolution blood smear images captured under various microscopic conditions. The dataset exhibits a multi-level hierarchical organization with three distinct training folds (fold_0, fold_1, fold_2), each containing two biologically significant classes: *all* (representing lymphoblasts indicative of Acute Lymphoblastic Leukemia) and *hem* (representing healthy lymphocytes and other blood components).



Fig. 1: Example blood smear images from the dataset

1. Data Loading and Annotation

To ensure balanced computational requirements and fair comparison, each dataset (*all* and *hem* from each fold) was limited to 500 images. This resulted in a total of 3,000 images per pipeline (500 images $\times$ 2 classes $\times$ 3 folds), maintaining consistent data volume across experiments while preserving the original class distribution.

Binary class labels were assigned following medical convention: label 1 for *all* (leukemia-positive) and label 0 for *hem* (healthy control). This labeling strategy ensured consistent ground truth representation across all experimental conditions.

2. Dataset Partitioning Strategy

Images from all three folds were aggregated into a unified dataset. The combined dataset underwent stratified splitting using an 80:20 ratio for training and validation sets, respectively. Stratification preserved the original class distribution in both splits, mitigating potential sampling bias and ensuring representative learning and evaluation.

b. Comparative Pipeline Architecture

Two fundamentally distinct preprocessing pipelines were constructed, diverging only in their initial feature transformation approach while maintaining identical subsequent components for fair comparison.

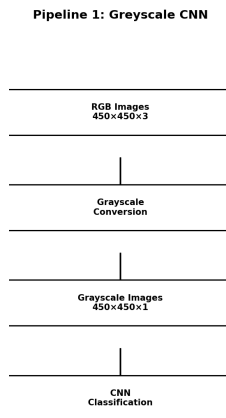


Fig. 2: Greyscale CNN pipeline architecture

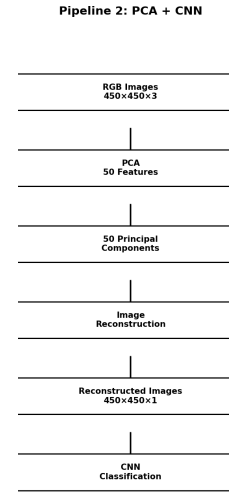


Fig. 3: PCA + CNN pipeline architecture

1. Pipeline 1: Greyscale CNN (Baseline Architecture)

As shown in Fig. 2, this pipeline implements a conventional computer vision preprocessing pathway with direct grayscale conversion and preservation of original spatial dimensions for CNN classification.

2. Pipeline 2: Principal Component Analysis + CNN

Fig. 3 illustrates the experimental pipeline incorporating dimensionality reduction through Principal Component Analysis. The mathematical transformation proceeds as follows:

Given a dataset of N RGB images, each image $I_i \in \mathbb{R}^{450 \times 450 \times 3}$ is first flattened into a vector $\mathbf{x}_i \in \mathbb{R}^D$ where $D = 450 \times 450 \times 3 = 607,500$. These vectors form the data matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times D}$.

The data is centered by subtracting the mean vector:

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i, \quad \mathbf{X}_{\text{centered}} = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^T \quad (1)$$

where $\mathbf{1}$ is an N -dimensional vector of ones.

The covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{D \times D}$ is computed and decomposed via Singular Value Decomposition (SVD):

$$\mathbf{\Sigma} = \frac{1}{N-1} \mathbf{X}_{\text{centered}}^T \mathbf{X}_{\text{centered}} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \quad (2)$$

where $\mathbf{V}$ contains the principal components (eigenvectors) and $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues.

The top $k = 50$ principal components corresponding to the largest eigenvalues are selected, forming the projection matrix $\mathbf{W} = \mathbf{V}_{:,k} \in \mathbb{R}^{D \times k}$. The original data is projected onto this reduced subspace:

$$\mathbf{Y} = \mathbf{X}_{\text{centered}} \mathbf{W} \in \mathbb{R}^{N \times k} \quad (3)$$

Reconstruction to the original space is performed as:

$$\hat{\mathbf{X}} = \mathbf{Y} \mathbf{W}^T + \mathbf{1}\bar{\mathbf{x}}^T \quad (4)$$

The reconstructed vectors $\hat{\mathbf{x}}_i$ are reshaped back to image format $\hat{I}_i \in \mathbb{R}^{450 \times 450 \times 3}$ and converted to grayscale for identical CNN processing as Pipeline 1.

c. Convolutional Neural Network Architecture

A consistent CNN architecture was implemented for both pipelines, featuring hierarchical feature learning through four sequential convolutional blocks with progressive downsampling:

- **Block 1:** 32 filters, reducing dimensions from 450×450 to 225×225
- **Block 2:** 64 filters, reducing to 112×112
- **Block 3:** 128 filters, reducing to 56×56
- **Block 4:** 256 filters, final reduction to 28×28

Each convolutional layer includes Batch Normalization and ReLU activation, with max-pooling for spatial reduction. The network concludes with global average pooling and fully-connected layers ($256 \rightarrow 512 \rightarrow 256 \rightarrow 2$) with Dropout regularization ($p = 0.5$), totaling approximately 3.2 million trainable parameters.

d. Training Protocol and Hyperparameter Configuration

Identical hyperparameters ensured fair comparison across pipelines:

- Adam optimizer with learning rate 0.001
- Cross-Entropy Loss for binary classification
- Batch size of 32 samples
- 25 training epochs
- Learning rate reduction by factor 0.1 every 10 epochs

e. Evaluation Framework

The analysis employed dual-focus evaluation assessing both classification performance and computational efficiency.

1. Classification Performance Metrics

- Accuracy, Precision, Recall, and F1-Score
- Training and validation loss trajectories
- Convergence behavior analysis

2. Computational Efficiency Metrics

- Training time and epoch duration
- Inference latency and throughput
- Memory footprint and resource utilization
- Parameter counts and model complexity

Custom instrumentation functions automatically logged all metrics, with comprehensive reports generated for precise comparisons between preprocessing methodologies.

III. RESULTS

a. Individual Pipeline Performance Analysis

1. Greyscale CNN Pipeline

The baseline greyscale pipeline demonstrated stable training characteristics with consistent convergence across 25 epochs. As shown in Table 1, the model achieved balanced performance between training and validation sets.

TABLE 1: GREYSCALE CNN PIPELINE PERFORMANCE METRICS

Metric	Train	Val
Final Accuracy	78.54%	78.50%
Final Loss	0.4944	0.5000
Accuracy Gap	0.04%	
Training Time	16369.79s	
Epoch Time	654.79s	

The training dynamics revealed minimal overfitting, with only 0.04% difference between training and validation accuracy. The loss curves showed stable convergence behavior throughout the training process.

2. PCA + CNN Pipeline

The PCA preprocessing pipeline exhibited distinct training characteristics with significant performance disparities. Table 2 summarizes the key metrics.

TABLE 2: PCA + CNN PIPELINE PERFORMANCE METRICS

Metric	Train	Val
Final Accuracy	97.58%	63.33%
Final Loss	0.0645	1.1690
Accuracy Gap	34.25%	
Training Time	16708.08s	
Epoch Time	668.32s	

This pipeline demonstrated severe overfitting, achieving near-perfect training accuracy (97.58%) but poor validation performance (63.33%). The validation loss showed significant instability throughout training, with multiple collapses to near-random performance levels.

b. Computational Efficiency Analysis

1. Computational Requirements by Pipeline

The FLOPS analysis revealed substantial computational demands for the PCA preprocessing stage. Table 3 details the computational requirements.

TABLE 3: COMPUTATIONAL REQUIREMENTS ANALYSIS

Component	Greyscale	PCA
Model Parameters	1,437,026	1,437,026
Model Size	5.48 MB	5.48 MB
Inference Throughput	12.52 img/s	11.17 img/s
Inference Latency	2556.12 ms	2864.37 ms
PCA SVD FLOPS	-	3.73e17
Total FLOPS	-	3.73e17

The PCA preprocessing introduced massive computational overhead, with the SVD operation alone requiring 373.5 quadrillion FLOPS. This represents the dominant computational cost in the PCA pipeline.

c. Comparative Performance Analysis

TABLE 4: COMPARATIVE PIPELINE PERFORMANCE SUMMARY

Metric	Grayscale	PCA	Diff.
Val Accuracy	78.50%	63.33%	-15.17%
Stability	High	Low	-
Overfitting	Minimal	Severe	-
Train Time	16369.79s	16708.08s	+2.07%
Inference Speed	12.52 img/s	11.17 img/s	-10.78%
Comp. Cost	Low	Very High	-

The grayscale pipeline outperformed the PCA approach across all key metrics, achieving superior validation accuracy with significantly lower computational requirements. The PCA pipeline's computational overhead, primarily from the SVD operation, did not translate into improved classification performance.

d. Training Dynamics Visualization

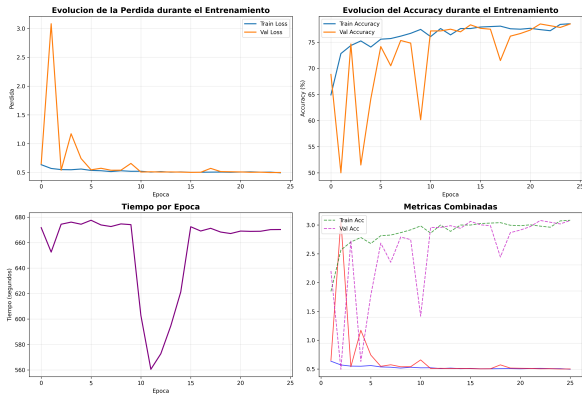


Fig. 4: Greyscale CNN training metrics

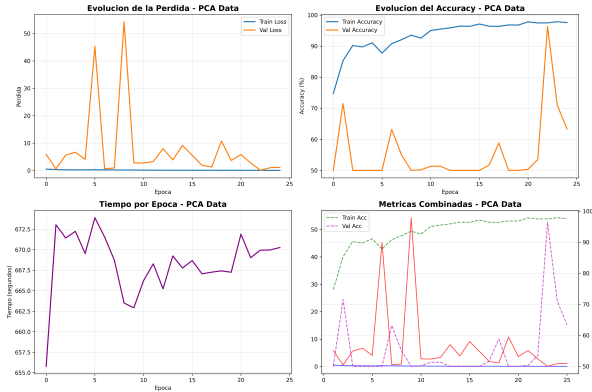


Fig. 5: PCA CNN training metrics

Fig. 4 and Fig.5 show the contrasting training behaviors between pipelines. The grayscale model maintained consistent validation performance, while the PCA model exhibited unstable validation metrics with multiple performance

collapses throughout training, indicating poor generalization capability despite excellent training performance.

IV. DISCUSSION

a. Individual Pipeline Analysis

1. Greyscale CNN Pipeline Performance

The greyscale pipeline demonstrated robust performance characteristics suitable for medical image classification. The minimal accuracy gap (0.04%) between training and validation sets indicates excellent generalization capability. This stability suggests that the direct grayscale conversion preserves sufficient discriminative features for leukemia classification while maintaining computational efficiency.

The model achieved 78.50% validation accuracy, which, while below clinical deployment standards, provides a solid baseline for comparison. The consistent loss reduction throughout training indicates proper convergence without overfitting concerns.

2. PCA + CNN Pipeline Challenges

The PCA pipeline exhibited severe overfitting with a 34.25% accuracy gap between training and validation performance. This suggests that the 50-component PCA reconstruction created features that were overly optimized for the training set but lacked generalizability. The multiple validation performance collapses observed during training indicate fundamental instability in the learned representations.

The excellent training accuracy (97.58%) demonstrates that the PCA-reconstructed images contain sufficient information for the model to memorize training examples, but the reconstructed features may not capture the essential pathological characteristics needed for generalization to unseen data.

b. Computational Efficiency Analysis

1. PCA Preprocessing Overhead

The FLOPS analysis reveals that the PCA preprocessing stage introduces prohibitive computational costs. The SVD operation alone requires 373.5 quadrillion FLOPS, dwarfing the actual CNN inference costs. This massive computational requirement makes the PCA approach impractical for real-time clinical applications or large-scale deployment.

The 2.07% increase in training time and 10.78% reduction in inference speed further compound the computational disadvantages of the PCA pipeline without providing corresponding performance benefits.

c. Comparative Analysis and Trade-offs

The experimental results clearly demonstrate that the theoretical benefits of PCA for dimensionality reduction do not translate to practical advantages in this leukemia classification context. The grayscale pipeline outperforms the PCA approach in both classification accuracy (78.50% vs. 63.33%) and computational efficiency.

The trade-off analysis reveals that the aggressive dimensionality reduction to 50 components likely discards critical discriminative information present in the original images. Medical images, particularly those used for leukemia classification, often contain subtle pathological features that may be lost during aggressive PCA reconstruction.

d. Limitations and Implications

The poor performance of the PCA pipeline challenges conventional assumptions about dimensionality reduction benefits in medical image analysis. While PCA can be effective for noise reduction and computational optimization in some domains, our results suggest that for fine-grained medical image classification tasks, preserving the original image structure through simple grayscale conversion provides superior results.

The computational overhead of PCA preprocessing, particularly the SVD operation, makes this approach unsuitable for practical clinical applications where both accuracy and efficiency are critical considerations.

V. CONCLUSION

This study demonstrates that for leukemia classification using CNN architectures, the greyscale preprocessing pipeline outperforms the PCA-based approach in both performance and efficiency. The greyscale model achieved 78.50% validation accuracy with minimal overfitting, while the PCA pipeline suffered from severe overfitting (63.33% validation accuracy) despite 97.58% training accuracy.

The massive computational overhead of PCA preprocessing (373.5 quadrillion FLOPS for SVD operations) further limits its practical utility. These findings indicate that for medical image classification requiring preservation of subtle pathological features, direct grayscale conversion provides a more effective and efficient preprocessing strategy than aggressive dimensionality reduction via PCA.

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